# Advanced Programming Paradigms

Project Assignment

# Project assignment

- Demonstrate knowledge of Advanced Programming Paradigms in practice by improving your Advanced Application Development Based on Development Templates (AADBDT) project
- You can demonstrate it on any other project, but the project specification needs to be verified

- **Describe programming testing concepts** (Outcome**: O1 – 10 points**)
  - Minimal: describe basic concepts as a part of project presentation
  - Preferred: describe advanced concepts as a part of project presentation

- **Add unit, integration, UI tests** (Outcome: **O2 – 10 points**, **O3 – 10 points**)
  - Minimal: cover 2 of these categories with 5 tests in total, and compare them with examples on the solution
  - Preferred: cover all these categories with at least 10 tests in total (the tests need to be implemented on architecturally at least 2 different parts), and compare them with examples on the solution

- **Describe testing and improving programming solution** (Outcome**: O4 – 10 points**)
  - Minimal: describe testing programming system and making it compatible with relevant forms as a part of project presentation
  - Preferred: offer solutions for decreasing time spent and allocated memory for completing given programming solution as a part of project presentation

- **Add aspects using aspect-oriented programming** (Outcome: **O6 – 12 points**)
  - Minimal: an aspect with 2 usages
  - Preferred: at least 2 aspects with at least 5 usages in total

- **Demonstrate version control workflow you use when implementing features or refactoring** (Outcome: **O7 – 12 points**)
  - Minimal: the solution should be available on a Git repository, and describe the strategy of source code management as a part of project presentation
  - Preferred: at least 2 branches on a Git repository, and describe the strategy of source code management as a part of project presentation

- **Apply refactoring best practices to improve your existing code** (Outcome: **O8 – 8 points**)
  - Minimal: show examples of compliance with 2 principles from the SOLID
  - Preferred: show examples of compliance with at least 4 principles from the SOLID

- **Add metrics and demonstrate how you can monitor your application health and performance** (Outcome: **O9 – 12 points**)
  - Minimal: 2 metrics which can be monitored
  - Preferred: at least 5 metrics, from which at least one should be metric created by you

- **Reduce coupling in code by using functional programming paradigm** (Outcome: **O5 – 12 points**)
  - Minimal: 2 different examples (different methods)
  - Preferred: at least 5 different examples (different methods)

- **Demonstrate how to package and run your app as a container** (Outcome: **O8 – 4 points**)
  - Preferred: the app should be containerized using Docker